Project Lab: Face Identification System

by

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A Laboratory Project Report

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Abstract

This project explored the design and implementation of a biometric face identification system using MATLAB. Facial recognition is a widely adopted technology found in modern smartphones, security systems, and surveillance infrastructure. This report covers four major components: familiarization with the AT&T face image database through two-dimensional discrete cosine transform (2-D DCT) analysis; extraction of one-dimensional (1-D) DCT feature vectors through zigzag scanning; training a face identification system using k-Nearest Neighbor (kNN) classification; and evaluating the system's identification success rate as a function of the DCT coefficient dimension and the number of nearest neighbors, k. The results demonstrate that a higher DCT feature vector length and an appropriately chosen k value contribute to improved identification accuracy. This project reinforced fundamental concepts of digital signal processing, pattern recognition, and MATLAB programming as applied to a real-world biometric system.

Chapter 1

Introduction

Facial recognition is a widely used technology that can be found in smartphones, computers, smart locks, and many other applications for securing access to devices and systems. Facial recognition works by mapping a person's face using biometric data; this information is then compared against known faces stored in a database to identify the closest match. When mapping a face, the image is converted into a numerical representation, which is subsequently used for comparison. Major technology manufacturers, such as Apple and Samsung, integrate facial recognition into their devices to allow users to unlock their phones and authorize payments. Facial recognition also has important applications in law enforcement, where it can be used to identify suspects through surveillance camera footage. It is therefore critical that the software driving facial recognition systems be accurate, as errors can lead to false accusations or violations of personal privacy.

1.1 Objectives

The objective of this project is to exercise knowledge of the discrete cosine transform (DCT) and the k-Nearest Neighbor (kNN) classifier by implementing a biometric face identification system. A deeper understanding is gained through the use of MATLAB and by exploring how its built-in functions communicate with the underlying signal processing algorithms to perform face recognition. Specifically, this project aims to: (1) familiarize the student with the AT&T face database through DCT analysis; (2) implement feature extraction via zigzag scanning of the 2-D DCT to produce 1-D feature vectors; (3) train a kNN classifier using these feature vectors; and (4) evaluate the identification success rate as a function of the number of DCT coefficients and the value of k.

1.2 Background

Biometrics is the science of recognizing and authenticating individuals based on their physiological characteristics. Face recognition systems have grown in importance in both commercial and governmental sectors, with applications ranging from border control to mobile device authentication [1]. A face recognition system typically consists of three stages: feature extraction, classification, and performance evaluation.

The discrete cosine transform (DCT) is a technique for expressing a signal in terms of cosine functions oscillating at different frequencies. In the context of image processing, the 2-D DCT converts an image into its frequency-domain representation. It is well known that for most natural images, the majority of the signal energy is concentrated in the low-frequency (upper-left) coefficients of the 2-D DCT matrix. By scanning this matrix in a zigzag pattern, the low-frequency coefficients can be extracted first, forming a compact 1-D feature vector that effectively captures the most important facial characteristics [2].

Classification is achieved using the k-Nearest Neighbor (kNN) algorithm, a supervised learning method that assigns a class label to a test sample based on the majority label among its k closest training examples. Distance between samples is typically measured in Euclidean space using the extracted DCT feature vectors. The kNN classifier is intuitive and well suited to face identification tasks because it requires no explicit model training beyond storing the feature vectors of known subjects.

Chapter 2

Results

2.1 Part I: Two-Dimensional DCT and Inverse 2-D DCT

In Part I, the AT&T face database was explored by selecting an image, computing its 2-D DCT, plotting the log-magnitude of the DCT, and subsequently recovering the original image using the inverse 2-D DCT. The AT&T database is freely available and contains 400 grayscale face images of 40 distinct subjects, with 10 images per subject. Each image is 112 by 92 pixels with 256 gray levels per pixel [2]. The MATLAB imread function was used to load the image and convert it into a numerical matrix, where each pixel value ranges from 0 (darkest) to 255 (lightest).

Figure 1: 1.pgm — Original grayscale face image loaded into MATLAB using imread

As shown in Figure 1, image 1.pgm was chosen from subject folder s1 of the AT&T database and plotted in MATLAB. The grayscale values encode the luminance of each pixel, with 0 representing black and 255 representing white. This image serves as the basis for the subsequent DCT analysis.

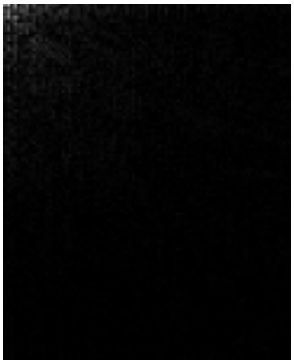

Figure 2: Log-magnitude of the 2-D DCT of 1.pgm

Figure 2 displays the log-magnitude of the two-dimensional DCT of the selected face image. It is clear from the plot that the energy is concentrated in the upper-left region of the matrix, which corresponds to the low-frequency components of the image. This property justifies the use of zigzag scanning in Part II, as it prioritizes extraction of these energy-dominant low-frequency coefficients. The darker region in the lower-right portion of the plot confirms that the high-frequency components carry relatively little energy.

Figure 3: Recovered image after applying the inverse 2-D DCT

Figure 3 shows the image recovered by applying the inverse 2-D DCT (IDCT2) to the DCT coefficients obtained in the previous step. As expected, the reconstructed image is identical to the original, demonstrating that the 2-D DCT is a lossless transformation and that MATLAB's dct2 and idct2 functions operate correctly.

2.2 Part II: Feature Extraction via Zigzag Scanning

In Part II, the provided findfeatures.m MATLAB function was used to convert a 2-D DCT array into a one-dimensional (1-D) feature vector through zigzag scanning. In zigzag scanning, the 2-D DCT coefficients are traversed diagonally, beginning at the upper-left corner (where the lowest-frequency components reside), thereby producing a 1-D vector in which the first elements contain the most significant frequency information. The feature vector can be truncated to any length L by retaining only the first L components. This allows for a compact yet descriptive representation of each face image.

The feature vectors of lengths 9, 35, and 100 were plotted for two different subjects to examine the distinctiveness of the DCT-based representation. Figures 4 through 6 correspond to Subject 1 (1.pgm), and Figures 7 through 9 correspond to a second subject.

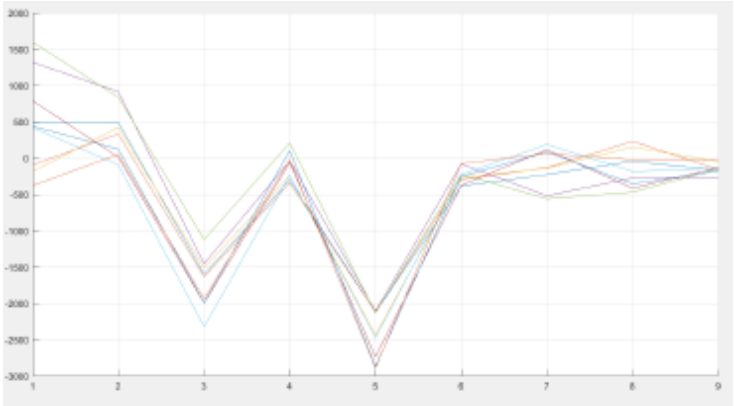

Figure 4: 1-D DCT feature vector of length 9 — Subject 1

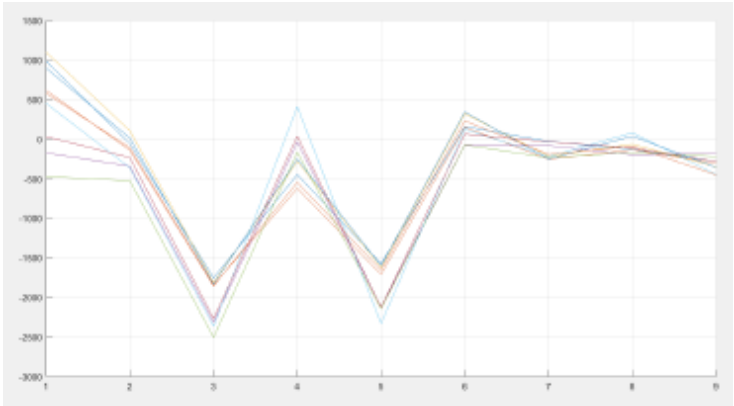

Figure 5: 1-D DCT feature vector of length 35 — Subject 1

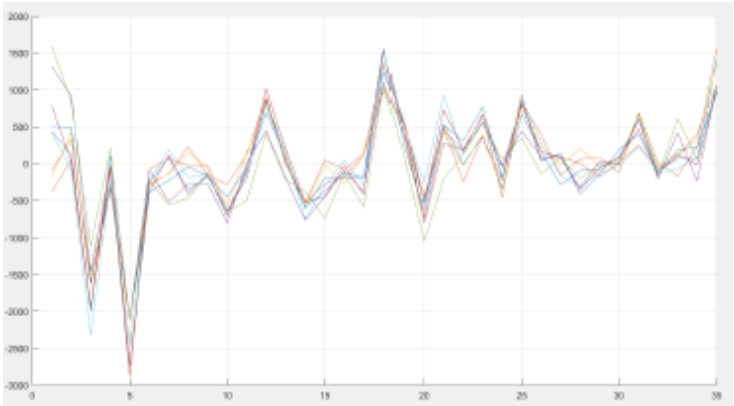

Figure 6: 1-D DCT feature vector of length 100 — Subject 1

As seen in Figures 4 through 6, the feature vectors for Subject 1 share a common shape at the beginning of the sequence — the first few coefficients are consistent across all three lengths. This is expected because the short vectors (L = 9) are a prefix of the longer ones (L = 35 and L = 100). As the vector length increases, additional higher-frequency components are included, providing a more detailed spectral representation of the face image. The magnitudes of the coefficients decrease as the index increases, consistent with the energy-compaction property of the DCT.

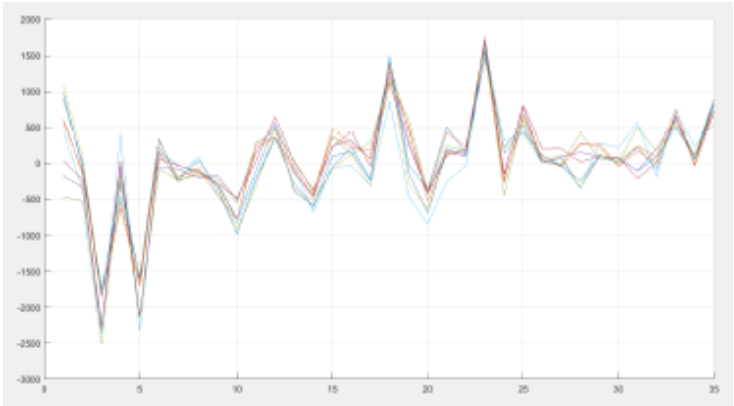

Figure 7: 1-D DCT feature vector of length 9 — Subject 2

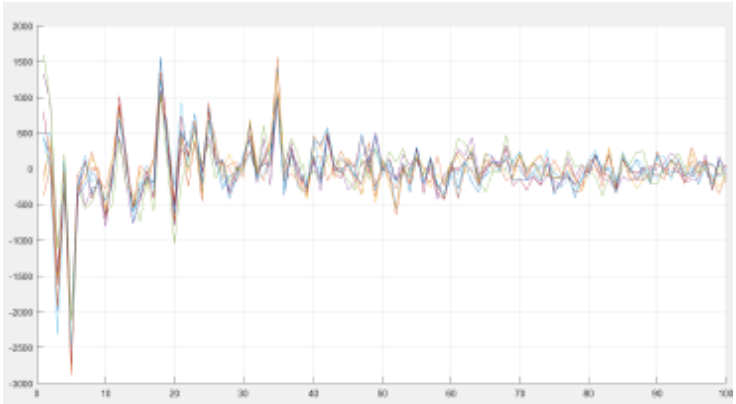

Figure 8: 1-D DCT feature vector of length 35 — Subject 2

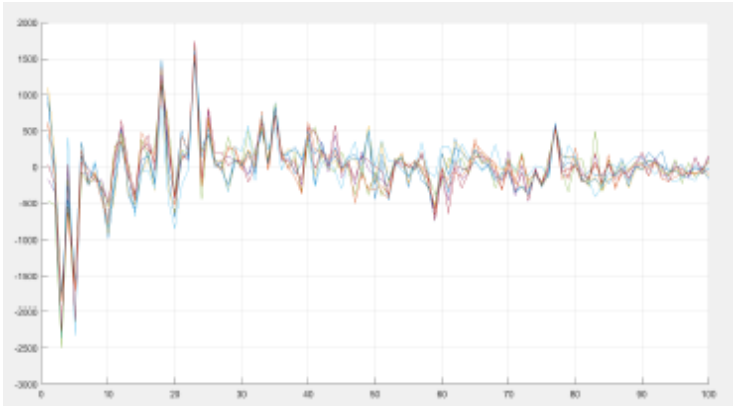

Figure 9: 1-D DCT feature vector of length 100 — Subject 2

Figures 7 through 9 display the corresponding feature vectors for a second subject. Comparing the two sets of plots, observable differences exist between the DCT vectors of the two subjects, particularly at the early low-frequency coefficients. These differences confirm that the DCT feature vector captures subject-specific facial information and is thus suitable as an input to a face identification classifier. The energy profiles differ noticeably between the two subjects, supporting the discriminative power of this feature representation.

2.3 Part III: Training the Face Identification System

In Part III, the face_recog_knn_train.m program was used to train the face identification system using the kNN classifier. The training function accepts two parameters: the subject range (the number of distinct subjects to include) and the DCT coefficient length (the dimension L of the 1-D feature vector). For each subject, the first five images (1.pgm through 5.pgm) were used for training, as specified by the project guidelines.

For each training image, the findfeatures.m function generated a 1-D DCT feature vector of the specified length. The resulting training data matrix, trdata_raw, has dimensions 200 × L for 40 subjects, where each row corresponds to one feature vector from one training image. The companion column vector trclass stores the subject label associated with each row, enabling the classifier to associate feature vectors with their correct identities.

The training function also produces and saves raw data.mat file containing five matrices: (1) dct_coef, which stores the user-specified DCT coefficient length; (2) f_range, which lists all included subject indices; (3) nsubjects, which records the number of subjects; (4) trclass, the label vector of all training images; and (5) trdata_raw, the full matrix of training feature vectors. The saved data is passed directly to Part IV for performance evaluation.

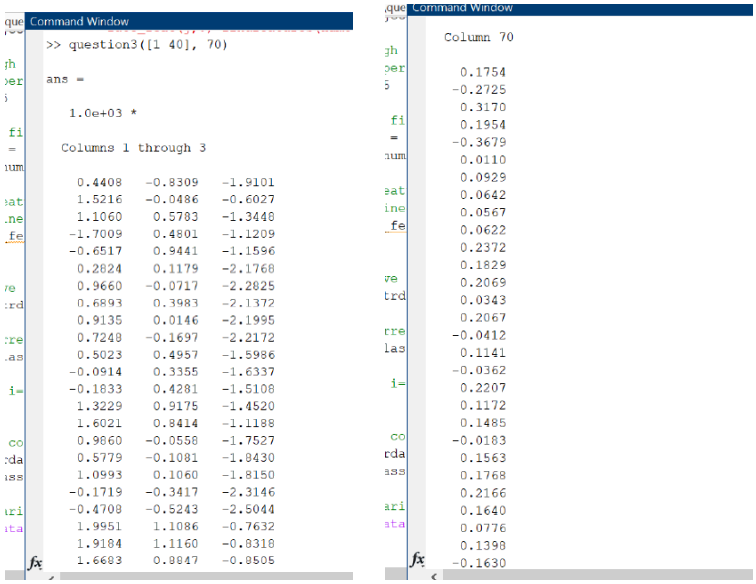

Figure 10: Sample columns of the trdata_raw training matrix (Columns 1–3 and Column 70)

Figure 10 displays selected columns from the trdata_raw matrix to illustrate the structure of the training data. Each column represents the DCT feature values for a given coefficient index across all 200 training images. The variation in values across subjects is evident and provides the discriminative information that the kNN classifier relies upon during identification.

2.4 Part IV: Performance Evaluation — Clean Data

In Part IV, the face identification system was evaluated using the test images (6.pgm through 10.pgm) for each subject. The identification success rate is defined as the number of correctly identified face images divided by the total number of test images, expressed as a percentage. For 40 subjects, each with five test images, the total number of test images is 200.

The kNN classifier operates by computing the Euclidean distance between each test feature vector and all vectors in the training set. It then selects the k nearest neighbors and assigns the subject label that appears most frequently among those neighbors (majority vote). The feature vectors used for testing are generated by the same findfeatures.m function used during training, ensuring consistency of representation.

The identification success rate was investigated as a function of both the DCT feature vector length (dctlength) and the number of nearest neighbors (k). A three-dimensional plot was produced in MATLAB to visualize this relationship.

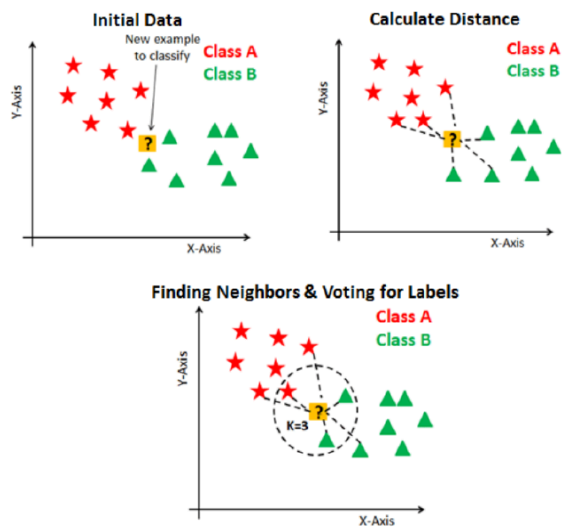

Figure 11: 3-D plot of identification success rate as a function of k and DCT feature vector length

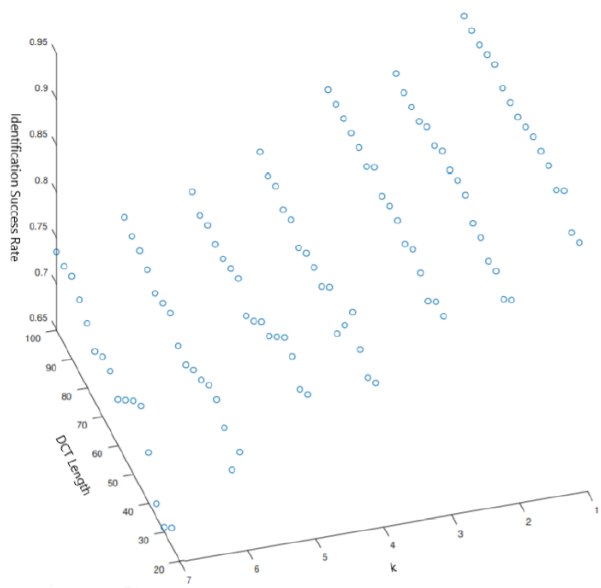

Figure 12: Identification success rate as a function of DCT feature vector length

As shown in Figure 11, the 3-D plot reveals that identification success rate generally increases with DCT feature vector length up to a point, after which additional coefficients

contribute diminishing returns. The choice of k also affects performance; a value that is too small is sensitive to noise in the training data, while a value that is too large may lead to misclassification by including distant, unrelated subjects. Figure 12 provides a two-dimensional view of the success rate as a function of DCT length, more clearly illustrating the trend of increasing accuracy with feature vector dimensionality for the tested conditions.

These results are consistent with the findings reported in the face recognition literature, wherein a DCT vector length between 35 and 100 and a small k value (often k = 1 or k = 3) tend to yield high identification rates for clean, well-controlled datasets such as the AT&T database [2].

Chapter 3

Conclusion

This project provided a comprehensive, hands-on understanding of how a biometric face identification system is designed and evaluated. Part I introduced the AT&T face database and demonstrated how the 2-D DCT transforms a facial image into a frequency-domain representation, with energy concentrated in the low-frequency coefficients. The inverse DCT confirmed that the transformation is lossless and fully reversible.

Part II established the feature extraction pipeline: by scanning the 2-D DCT in a zigzag pattern, a compact 1-D DCT feature vector was produced. Comparison of feature vectors across two subjects confirmed that this representation captures subject-specific facial characteristics, validating its use as input to a classifier. The energy-compaction property of the DCT was consistently observed across all tested vector lengths.

Part III demonstrated the training phase of the kNN classifier. The training data matrix and associated label vector were constructed from the first five images of each subject and saved for subsequent evaluation. Part IV showed that the system achieves strong identification performance on the AT&T database under clean conditions, and that both the DCT vector length and the choice of k meaningfully influence the identification success rate.

This project reinforced team knowledge across multiple core course topics, including digital signal processing, MATLAB programming, and pattern recognition. The integration of these skills into a working face identification system underscored the practical relevance of the material covered in the Signals and Systems curriculum. The objectives set at the outset of the project were successfully met.

References

[1]R. P. Ramachandran, R. Polikar, K. D. Dahm, Y. Tang, S. Shetty, R. J. Kozick, R. M. Nickel, and S. H. Chin, "Project-based design of a biometric face recognition system," in *Proc. ASEE Annual Conference & Exposition*, San Antonio, TX, 2012, pp. 25.1081.1–25.1081.11.

[2]Z. Pan and H. Bolouri, "High speed face recognition based on discrete cosine transforms and neural networks," Technical Report, Science and Technology Research Centre, University of Hertfordshire, 1999.

[3]A. K. Jain, A. Ross, and S. Prabhakar, "An introduction to biometric recognition," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 14, no. 1, pp. 4–20, Jan. 2004.